

QuadMorph: Material-Enabled Reconfigurable Shins for Wheel-Legged Quadrupedal Locomotion

Abstract—Wheeled locomotion provides high efficiency on smooth terrain, while legged locomotion enables traversal of complex environments. Combining these capabilities, however, typically requires additional actuators or mechanically complex wheel-leg mechanisms. We present *QuadMorph*, a quadrupedal robot platform with material-enabled reconfigurable shins that transform between wheel-like and leg-like geometries. Each shin integrates a custom 3D-printed thermally responsive polymer with liquid-metal Joule heating. In its stiff state, the shin maintains a circular geometry for wheeled locomotion. Heating softens the polymer, allowing the robot to exploit its existing joint actuators and environmental contacts to reshape the shin into a leg-like structure, which regains rigidity after cooling. This enables wheel-to-leg reconfiguration without dedicated morphing actuators. We develop reinforcement-learning policies for locomotion in both morphologies and validate the system on an open-source quadruped. Experiments characterize the material and reconfiguration behavior and demonstrate wheeled and legged locomotion, showing how programmable material properties can enable mechanically simple, multi-modal mobility.

I. INTRODUCTION

Legged robots have demonstrated remarkable mobility over complex and unstructured terrain, enabled by advances in mechanical design, control, and reinforcement learning (RL) [1], [2], [3], [4], [5]. However, legged locomotion remains energetically costly when traversing large and smooth environments where discrete footholds are unnecessary. Wheeled locomotion provides complementary advantages, offering high speed and efficiency on relatively flat surfaces. These complementary characteristics have motivated the development of wheel-legged robots that combine the terrain adaptability of legs with the efficiency of wheels [6], [7], [8].

Wheel-legged robots have demonstrated impressive capabilities ranging from high-speed rolling to dynamic locomotion over challenging terrain [6], [7]. Nevertheless, most existing wheel-legged quadrupeds retain wheels as permanent components of the robot. Other systems employ transformable wheel-leg mechanisms that change their contact geometry to negotiate different terrains [9], [10]. Such transformations, however, are typically realized through dedicated mechanical mechanisms, introducing additional components and mechanical complexity. More fundamentally, in conventional wheel-legged quadrupeds, the wheel remains a wheel even when the surrounding leg changes its configuration. This raises an intriguing question: *can the physical structure that serves as a wheel itself transform into a leg?*

In this work, we explore an alternative approach based on

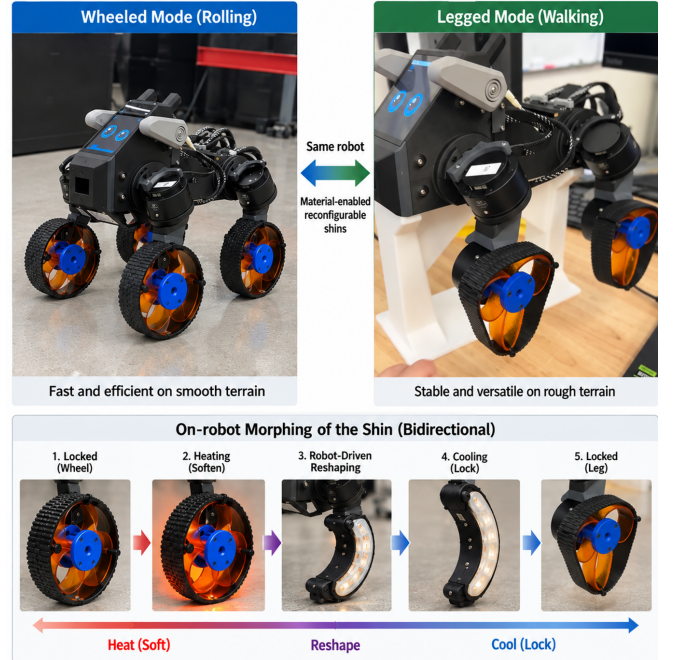


Fig. 1: **QuadMorph**: A quadruped with material-enabled reconfigurable shins for wheeled and legged locomotion.

material-enabled morphological reconfiguration. Variable-stiffness materials and structures have been investigated in soft and adaptive robotics to enable transitions between compliant and load-bearing states [11], [12]. In particular, thermally responsive polymers can undergo substantial changes in stiffness with temperature, enabling deformation in a compliant state followed by structural locking after cooling [11]. Rather than using this property solely for soft actuation or variable compliance, we use it to temporarily unlock the geometry of a load-bearing component of a legged robot. The robot can then use its existing joint actuators to reshape its own body. This separates *morphological activation* from *mechanical actuation*: thermal energy changes the mechanical properties of the structure, while the robot’s existing motors provide the forces required for reconfiguration.

We realize this concept in **QuadMorph**, an open-source quadrupedal robot equipped with material-enabled reconfigurable shins that transform between wheel-like and leg-like geometries. Each shin consists of deformable segments fabricated from a custom 3D-printed thermally responsive polymer and integrated with liquid-metal Joule heating. At normal operating temperature, the material is stiff and

maintains a circular geometry, allowing the shin to function as a wheel. Heating softens the polymer, allowing the robot to use its existing leg actuators and environmental contacts to selectively deform the shin into a leg-like structure. Subsequent cooling restores its rigidity and locks the new morphology. Reversing this process recovers the wheel geometry. Importantly, the transformation requires no dedicated morphing actuator.

The resulting platform supports two fundamentally different modes of locomotion using the same distal limb structure. In wheel mode, continuous rolling contact enables efficient locomotion over smooth terrain. In leg mode, discrete ground contacts enable conventional quadrupedal locomotion and traversal of terrain unsuitable for rolling. We develop RL-based controllers for both morphologies and experimentally demonstrate the resulting wheeled and legged locomotion capabilities.

The main contributions of this work are:

- A **material-enabled reconfigurable shin** that transforms between wheel-like and leg-like geometries, allowing the same physical structure to support two distinct modes of locomotion;
- An **actuator-reusing reconfiguration mechanism** in which liquid-metal Joule heating softens a custom 3D-printed thermally responsive polymer, enabling the robot's existing leg actuators and environmental contacts to reshape its morphology without dedicated morphing actuators;
- A complete quadrupedal system with **RL-based wheeled and legged locomotion**, together with experimental characterization of the material, reconfiguration process, and locomotion capabilities.

II. RELATED WORK

A. Wheel-Legged and Reconfigurable Locomotion

Wheels and legs provide complementary mobility characteristics, motivating robotic systems that combine the efficiency of rolling with the terrain adaptability of legged locomotion. One common approach is to attach actuated wheels to articulated legs. Wheeled-legged platforms such as ANYmal with actuated wheels have demonstrated efficient rolling, dynamic whole-body motions, and traversal of challenging terrain [6], [7]. Similar hybrid designs have been explored on other robotic platforms [8]. These systems retain the articulation of conventional legs while introducing wheels at the distal links, enabling both rolling and stepping behaviors. However, the wheels remain permanent elements of the robot and retain a fixed contact geometry across locomotion modes.

An alternative direction is to physically reconfigure the wheel itself. Sun et al. [9] proposed a transformable wheel-legged mechanism that switches between circular wheels for efficient rolling and an opened configuration for obstacle traversal. The Land Devil Ray (LDR) robot introduced passive and active transformation strategies based on a multi-link mechanism, reducing the need for additional transformation actuation [15]. FUHAR employs a four-bar-based

transformation mechanism to switch between circular wheels and multi-segment wheel-legs [16]. More recently, Fu et al. [10] developed a rim-shape-changeable wheel whose geometry and driving four-bar mechanism are jointly designed for smooth obstacle traversal. These systems demonstrate that actively changing the contact geometry can substantially expand the mobility of wheeled robots.

Table I summarizes representative approaches to combining wheeled and legged mobility. Conventional wheeled-legged quadrupeds achieve multimodal locomotion while maintaining a fixed wheel geometry, whereas transformable wheel-leg systems introduce mechanical structures to reconfigure the contact geometry. In contrast, QuadMorph realizes geometric reconfiguration through the load-bearing material itself.

Existing transformable wheel-leg systems predominantly realize morphological change through rigid-body mechanisms such as linkages, joints, triggers, and clutches [9], [15], [16], [10]. QuadMorph takes a different approach: the load-bearing structure itself becomes reconfigurable. Thermal activation temporarily reduces the stiffness of the shin, allowing forces generated by the robot's existing leg actuators and environmental contacts to reshape its geometry. Cooling subsequently restores the structural stiffness and locks the new morphology. Thus, rather than adding a rigid kinematic mechanism for transformation, QuadMorph uses *programmable material stiffness as the reconfiguration mechanism*.

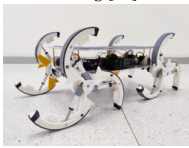
B. Smart Materials and Variable-Stiffness Robotics

Variable-stiffness materials provide robots with the ability to transition between compliant states suitable for deformation and rigid states suitable for load bearing. This capability has been extensively investigated in soft and adaptive robotics, where variable stiffness can improve force transmission, payload capacity, and interaction with the environment. Existing approaches exploit mechanisms including phase-transition materials, thermally responsive polymers, and low-melting-point metals [?].

Thermally responsive polymers are particularly attractive for robotic structures because their mechanical properties can change substantially across a transition temperature while remaining compatible with additive manufacturing. Recent work has integrated 3D-printed thermally responsive polymers with reconfigurable structures to achieve large deformation followed by structural locking [11]. Such systems demonstrate how material state transitions can provide both deformability during reconfiguration and structural support after the desired geometry is reached.

Liquid metals provide additional functionality due to their electrical conductivity and high deformability. They have been incorporated into soft robotic structures for variable stiffness and embedded Joule heating [12]. Because conductive liquid-metal paths can deform together with the surrounding structure, they provide a particularly attractive means of integrating thermal activation without introducing rigid heating components.

TABLE I: Comparison of representative wheeled-legged and reconfigurable quadruped robots.

System	ANYmal-W [6]	Skating Quadruped (Go1) [?]	DTransleg [13]	Swheg [14]	QuadMorph (Ours)
					
Wheel-Leg Morphing	× (Fixed wheels)	× (Interchangeable)	✓ (Shape-changing)	✓ (Transformable legs)	✓ (Material-enabled)
Contact Geometry	Fixed	Fixed	Reconfigurable	Reconfigurable	Reconfigurable
Reconfigurable Element	Rigid wheel module	Interchangeable distal unit	Circular wheel structure	Articulated leg mechanism	Load-bearing shin material
Morphing Mechanism	—	Manual replacement	Active mechanism	Mechanical linkage	Thermal softening
Morphing Actuation	—	—	Yes	Yes	No (uses existing joints)

QuadMorph builds upon these material capabilities but uses them for a different robotic function. Rather than primarily controlling compliance or using the smart material itself as an actuator, we use thermal softening to *temporarily unlock the geometry of a load-bearing robotic structure*. Embedded liquid metal provides Joule heating, while the mechanical work required for reconfiguration is supplied by the quadruped’s existing joint actuators through interaction with the environment. Cooling then restores the load-bearing stiffness of the structure. In this way, QuadMorph couples material-level stiffness modulation with robot-level actuation to achieve a discrete change in locomotion morphology, allowing the same distal structure to function as either a wheel or a leg.

III. QUADMORPH DESIGN

A. System Overview

QuadMorph is built upon an open-source quadrupedal robot (Pupper-v3 [17]) and replaces its conventional distal leg links with four reconfigurable shins. The design enables two distinct locomotion morphologies using the same physical structure, as illustrated in Fig. 1. In the *wheel mode*, each shin forms an approximately circular profile and serves as a load-bearing wheel, enabling the robot to locomote through continuous rolling contact. In the *leg mode*, the same structure is deformed into an elongated shin, providing a conventional distal link for quadrupedal locomotion.

The key design principle of QuadMorph is to separate *morphological activation* from *mechanical actuation*. Instead of introducing an additional actuator to mechanically transform the shin, we use a thermally responsive material whose stiffness can be actively modulated. Embedded liquid-metal channels provide Joule heating to temporarily soften the structure, while the robot’s existing joint actuators provide the mechanical forces required to reshape it. After reconfiguration, cooling restores the structural stiffness and mechanically locks the shin in the desired morphology. Consequently, electrical power is primarily required during reconfiguration rather than for maintaining either locomotion morphology.

B. Reconfigurable Shin Design

Each reconfigurable shin consists of three deformable segments arranged around the perimeter of the structure, as

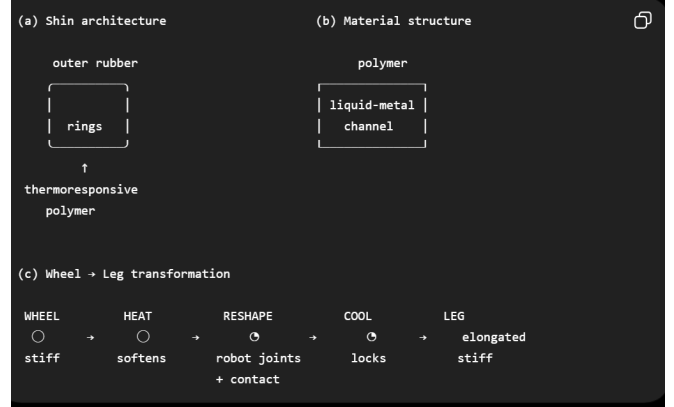


Fig. 2: Reconfigurable shin design and transformation principle.

shown in Fig. 2 (a, b). The segments are fabricated from a custom thermally responsive polymer and integrated into a compliant outer ring. Liquid-metal conductive channels are embedded within the deformable segments to provide distributed Joule heating. The outer elastomeric structure constrains the segments into the desired overall geometry while also providing a compliant contact surface during locomotion.

We denote the effective lengths of the three deformable segments as l_1, l_2 , and l_3 . In wheel mode, the three segments assume their nominal configuration, $l_i = l^N, i \in \{1, 2, 3\}$, forming an approximately circular profile with effective radius r_W . The resulting structure acts directly as the robot’s wheel, without requiring a separate wheel attached to the distal link.

To obtain the leg morphology, the geometry of the three segments is redistributed. One segment is elongated while the remaining two are compressed: $l_1 > l^N, l_2 \approx l_3 < l^N$, where the segment indices can be defined according to the orientation of the shin. This deformation collapses the circular profile into an elongated structure that functions as the distal link of the leg. Importantly, both configurations are supported by the same continuous physical structure; no wheel is attached, detached, or mechanically deployed during the transformation.

C. Robot-Driven Morphological Reconfiguration

Transformation between the two locomotion morphologies is achieved through a five-stage process:

Heat $\rightarrow$ Soften $\rightarrow$ Robot Reshape $\rightarrow$ Cool $\rightarrow$ Lock.

The process is illustrated in Fig. 2 (c).

In its nominal state, the thermally responsive polymer has sufficiently high stiffness to sustain the loads encountered during locomotion. To initiate reconfiguration, current is applied through the embedded liquid-metal channels. Joule heating raises the temperature of the polymer and reduces its stiffness, $P_h = I^2 R$, where I and R denote the applied current and electrical resistance of the embedded conductive path, respectively. Once the material reaches the reconfiguration temperature, the shin becomes sufficiently compliant to undergo large deformation under forces generated by the robot.

TUND the next two paragraphs are Not accurate. Rather than using a dedicated morphing actuator, QuadMorph exploits its existing leg joints together with contact against the environment to reshape the softened shin. During the wheel-to-leg transition, the robot places the shin against the ground and executes a prescribed joint motion that loads the deformable segments asymmetrically. This loading compresses two segments while extending the remaining segment, driving the transformation. Once the target geometry is reached, heating is removed while the robot maintains the configuration. As the material cools, its stiffness increases and the transformed geometry becomes load bearing. The robot can then remove the reconfiguration load and initiate legged locomotion.

The reverse procedure is used for the leg-to-wheel transition. The material is reheated, the robot applies the required contact forces to recover the approximately circular geometry, and the structure is subsequently cooled and locked in wheel mode. This mechanism allows the robot's existing actuators to serve a dual purpose: they generate locomotion in the locked configurations and provide the mechanical work for morphological reconfiguration when the shin is softened.

D. Thermally Reconfigurable Material

CLW check this subsection, add some details w The reconfigurable shin requires a material that combines high structural stiffness during locomotion with large deformability during reconfiguration. To achieve these competing requirements, we develop a custom 3D-printable thermally responsive polymer whose mechanical properties vary substantially with temperature. The polymer is fabricated using a custom resin formulation and printing procedure that enables the geometry and embedded conductive features required by the shin.

Conceptually, the temperature-dependent stiffness can be represented as

$$E(T) \approx \begin{cases} E_{\text{stiff}}, & T < T_{\text{tr}}, \\ E_{\text{soft}}, & T \geq T_{\text{tr}}, \end{cases} \quad E_{\text{stiff}} \gg E_{\text{soft}}, \quad (1)$$

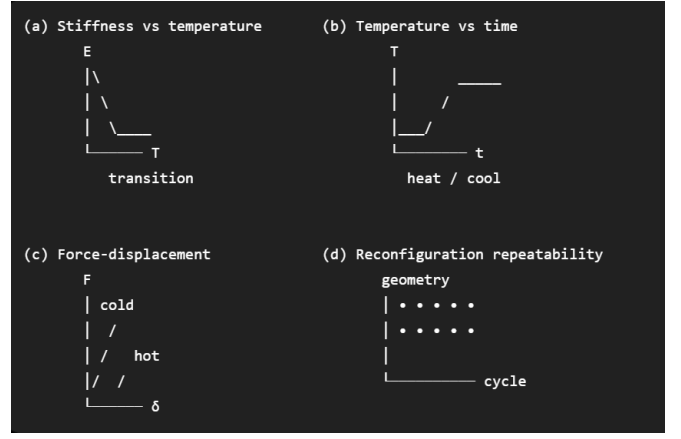


Fig. 3: Material and shin characterization.

where T_{tr} denotes the effective transition temperature. Below T_{tr} , the material provides the rigidity required to maintain the wheel or leg geometry under locomotion loads. Above T_{tr} , its reduced stiffness permits the large deformation required for morphological reconfiguration.

Liquid-metal channels embedded in the printed structure serve as compliant resistive heaters. Unlike conventional rigid heating elements, the conductive channels can deform together with the polymer during morphing while maintaining electrical continuity. This enables heating to be distributed directly throughout the regions undergoing reconfiguration and avoids introducing rigid components that would restrict deformation.

The material formulation and fabrication procedure are designed specifically around the mechanical requirements of QuadMorph. In Sec. V, we experimentally characterize the temperature-dependent mechanical response, heating and cooling dynamics, and cyclic reconfiguration performance of the fabricated structures.

IV. LOCOMOTION AND RECONFIGURATION CONTROL

TUND check the details

QuadMorph operates in two mechanically distinct configurations: a wheel mode with continuous rolling contact and a leg mode with discrete ground contacts. We train separate reinforcement learning (RL) policies for the two locomotion modes and use a state-machine-based controller to execute the physical reconfiguration between them. Both locomotion policies directly command the existing joint actuators of the quadruped; no additional actuator is introduced for either locomotion or morphological reconfiguration.

A. Reinforcement Learning Formulation

We formulate locomotion as a Markov decision process (MDP) and train the policies in simulation using proximal policy optimization (PPO). At each control step, the policy receives an observation $\mathbf{o}_t$ and outputs joint position targets

$$\mathbf{a}_t = \pi(\mathbf{o}_t), \quad \mathbf{q}_t^d = \mathbf{q}^0 + \mathbf{a}_t, \quad (2)$$

where $\mathbf{q}^0$ denotes the nominal joint configuration. The target positions are tracked by joint-level PD controllers,

$$\tau = \mathbf{K}_p(\mathbf{q}^d - \mathbf{q}) - \mathbf{K}_d\dot{\mathbf{q}}. \quad (3)$$

The policy observation follows a standard proprioceptive locomotion formulation,

$$\mathbf{o}_t = [\boldsymbol{\omega}_t, \mathbf{g}_t, \mathbf{v}_t^{\text{cmd}}, \mathbf{q}_t - \mathbf{q}^0, \dot{\mathbf{q}}_t, \mathbf{a}_{t-1}], \quad (4)$$

where $\boldsymbol{\omega}_t$ is the base angular velocity, $\mathbf{g}_t$ is the gravity vector expressed in the body frame, $\mathbf{v}_t^{\text{cmd}}$ contains the commanded planar velocity and yaw rate, and $\mathbf{a}_{t-1}$ is the previous action. The exact observation and reward terms used for each morphology are summarized in Table ??.

B. Wheeled Locomotion

In wheel mode, the circular shins form passive rolling interfaces with the environment. Unlike conventional wheeled-legged robots equipped with actuated wheels, QuadMorph contains no dedicated wheel motors. Forward motion must therefore be generated entirely through coordinated motion of the existing leg joints.

We train a dedicated wheeled locomotion policy, π_{wheel} , to discover joint motions that exploit the rolling contact of the circular shins. The primary task objective is velocity tracking,

$$r_{\text{vel}} = \exp\left(-\frac{\|\mathbf{v}_{xy} - \mathbf{v}_{xy}^{\text{cmd}}\|^2}{\sigma_v}\right), \quad (5)$$

together with yaw-rate tracking and regularization terms penalizing undesired body motion, excessive joint torques, and rapid changes in actions. The total reward is written as

$$r_t = w_v r_{\text{vel}} + w_\omega r_{\text{yaw}} + w_o r_{\text{ori}} - w_\tau p_\tau - w_a p_{\Delta a}. \quad (6)$$

The circular shin geometry and its contact properties are explicitly modeled in simulation. During training, we randomize relevant physical parameters, including ground friction, robot mass properties, motor strength, and shin contact properties, to improve robustness to the sim-to-real discrepancy introduced by the deformable physical structure.

Importantly, we do not prescribe a rolling gait or kinematic trajectory. The RL policy learns how to coordinate the articulated legs to generate and regulate rolling motion using the passive wheel-shaped shins.

C. Legged Locomotion

After morphological transformation, the elongated shins behave as conventional distal leg links and interact with the environment through discrete contacts. We therefore train the legged policy π_{leg} using a standard quadrupedal locomotion pipeline.

The policy tracks commanded planar velocity and yaw rate while maintaining base stability and regularizing joint motion and actuation effort. The reward follows the standard form

$$r_t^{\text{leg}} = w_v r_{\text{vel}} + w_\omega r_{\text{yaw}} + w_o r_{\text{ori}} + w_f r_{\text{feet}} - w_\tau p_\tau - w_a p_{\Delta a}, \quad (7)$$

where r_{feet} represents foot-contact-related objectives used to encourage stable stepping behavior.

Although the learning procedure follows a conventional legged locomotion pipeline, the simulation model incorporates the geometry and inertial properties of the transformed QuadMorph shins. Domain randomization is again applied to account for manufacturing variation and differences between the nominal and realized shin geometries.

D. Morphological Reconfiguration Control

Locomotion and morphological reconfiguration are coordinated using the finite-state machine illustrated in Fig. ??. The robot operates in one of five states,

$$\mathcal{M} = \{\text{Wheel, Heating, Reshaping, Cooling, Leg}\}. \quad (8)$$

During normal locomotion, the material remains in its stiff state and either π_{wheel} or π_{leg} controls the robot. When a morphology change is requested, locomotion is first terminated and the robot assumes a predefined reconfiguration pose. Electrical current is then applied to the liquid-metal heaters until the shin reaches the desired reconfiguration temperature.

Once softened, the locomotion policy is temporarily replaced by a prescribed joint-space reconfiguration trajectory,

$$\mathbf{q}^d(t) = \mathbf{q}_{\text{morph}}(t), \quad (9)$$

which uses contact with the environment to apply the forces required to reshape the shins. After the target geometry is reached, the joint configuration is maintained while heating is removed. The material cools and recovers its load-bearing stiffness, after which the corresponding locomotion policy is activated.

The overall control architecture can therefore be expressed as

$$\pi = \begin{cases} \pi_{\text{wheel}}, & m = \text{Wheel}, \\ \mathbf{q}_{\text{morph}}(t), & m = \text{Reshaping}, \\ \pi_{\text{leg}}, & m = \text{Leg}. \end{cases} \quad (10)$$

This separation between learned locomotion and deterministic reconfiguration simplifies policy training while allowing the same actuators to perform both locomotion and morphological transformation.

V. EXPERIMENTAL RESULTS

We evaluate QuadMorph through experiments at three levels: material characterization, component-level reconfiguration, and full-system locomotion. The experiments are designed to answer the following questions: (1) whether the proposed material provides sufficient temperature-dependent stiffness variation for both reconfiguration and load-bearing locomotion; (2) whether the fabricated shins can reliably transform between wheel and leg morphologies; and (3) whether the resulting robot can exploit both morphologies for wheeled and legged locomotion.

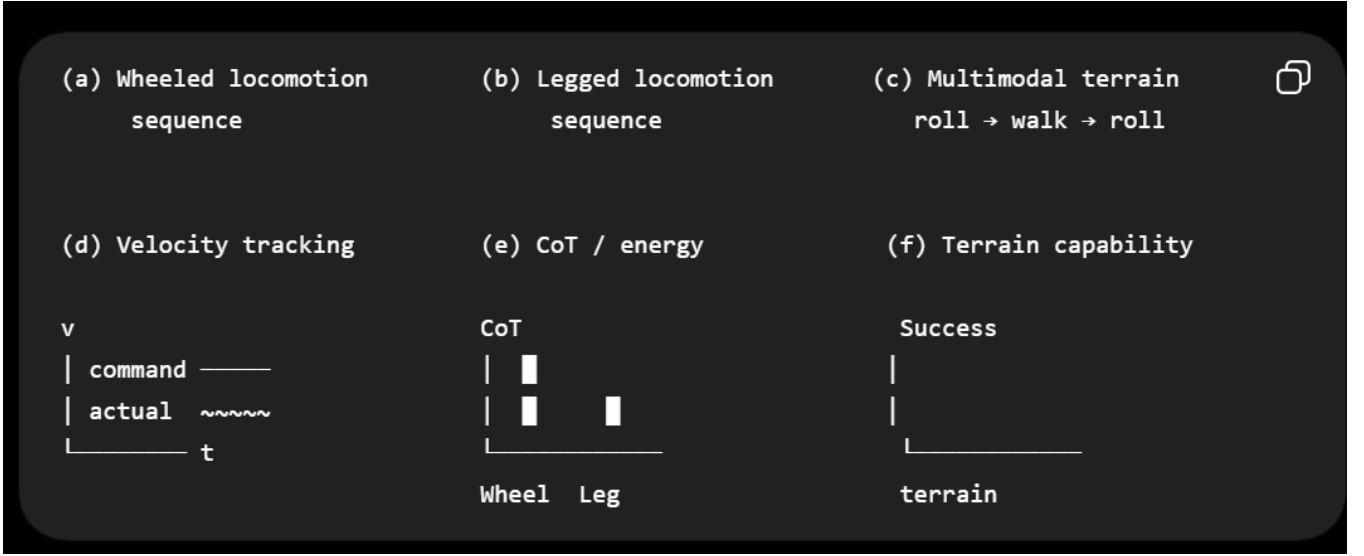


Fig. 4: Locomotion results: Rolling Montage (flat ground, up and down slopes), Legged Montage (flat ground, rubbles, grass, etc); Velocity tracking Comparison between legged and wheel (simulation & hardware); COT Comparison between legged and wheel (simulation & hardware);

A. Material Characterization

CLW check We first characterize the thermo-mechanical behavior of the custom 3D-printed polymer. Specimens are fabricated using the same material composition and printing procedure as the reconfigurable shins. Mechanical tests are performed across a range of temperatures to quantify the change in stiffness associated with thermal activation.

Fig. ??(a) shows the measured [Young’s modulus/bending stiffness] as a function of temperature. The material transitions from a stiff state of approximately $[E_{\text{cold}}]$ at room temperature to $[E_{\text{hot}}]$ at the reconfiguration temperature $[T_{\text{tr}}]$ °C, corresponding to a stiffness ratio of

$$\gamma_E = \frac{E_{\text{cold}}}{E_{\text{hot}}} = [\cdot]. \quad (11)$$

This large stiffness variation allows the same structure to resist locomotion loads in its cold state while remaining sufficiently compliant for robot-driven deformation after heating.

We next characterize the embedded liquid-metal heating system. Fig. ??(b) shows the temperature response under an applied electrical power of $[P]$ W. The material reaches the reconfiguration temperature in $[t_h]$ s and cools below the locking temperature in $[t_c]$ s after heating is removed. Thermal images confirm that the embedded liquid-metal channels provide distributed heating throughout the deformable region.

B. Shin Reconfiguration

We next evaluate whether the fabricated shin can repeatedly transform between its two load-bearing morphologies. Starting from wheel mode, the liquid-metal heater is activated until the polymer reaches its compliant state. External loading representative of the robot-driven reconfiguration process is then applied to transform the structure into the leg configuration. The structure is held in this configuration

during cooling, after which the external load is removed. The reverse procedure is used for leg-to-wheel transformation.

Fig. ?? shows the complete transformation sequence. The shin changes from an approximately circular profile with diameter $[D_W]$ to an elongated leg configuration with effective length $[L_L]$. The wheel-to-leg and leg-to-wheel transformations require $[t_{W \rightarrow L}]$ s and $[t_{L \rightarrow W}]$ s, respectively.

To quantify repeatability, we perform $[N]$ consecutive wheel–leg–wheel cycles. The resulting geometries exhibit an average [dimensional/reconfiguration] error of $[\cdot]$ across the trials. We further perform static loading tests in both locked configurations. The transformed shin withstands a maximum load of $[\cdot]$ N without structural failure, demonstrating sufficient load-bearing capability for quadrupedal locomotion.

C. Wheeled Locomotion

We evaluate the learned wheeled locomotion policy on the physical QuadMorph platform. In this configuration, all four shins are locked in their circular geometry and behave as passive rolling contacts. The robot receives commanded forward, lateral, and yaw velocities and generates locomotion exclusively through coordinated motion of its existing leg joints.

The learned policy successfully produces sustained rolling locomotion without dedicated wheel actuators. Fig. ??(a) shows the robot tracking commanded forward velocities ranging from $[v_{\text{min}}]$ to $[v_{\text{max}}]$ m/s. Across the tested commands, the mean velocity tracking error is $[\cdot]$ m/s. The robot additionally demonstrates turning and yaw-rate tracking, as shown in Fig. ??(b).

To quantify the benefit of the wheel morphology, we compare wheeled and legged locomotion over the same flat-ground course. We measure traversal time and electrical energy consumption and compute the cost of transport

$$\text{CoT} = \frac{E}{mgd}, \quad (12)$$

where E is the consumed energy, m is the robot mass, g is gravitational acceleration, and d is the traveled distance. QuadMorph in wheel mode achieves a CoT of $[\cdot]$, compared with $[\cdot]$ in leg mode, corresponding to a $[\cdot]\%$ reduction in locomotion energy on flat terrain.

D. Legged Locomotion

We next evaluate locomotion after transforming the shins into their elongated configuration. The transformed structures remain passive and load bearing during locomotion, and no heating is required to maintain the leg morphology.

Using the learned legged locomotion policy, QuadMorph successfully tracks velocity commands on flat ground and traverses uneven terrain. We evaluate the robot on [rough terrain, discrete obstacles, slopes, and/or stairs], as illustrated in Fig. ???. The robot reaches a maximum forward walking velocity of $[\cdot]$ m/s and traverses obstacles up to $[\cdot]$ cm in height.

These experiments demonstrate that the reconfigurable shin is not merely capable of assuming a leg-like geometry, but provides sufficient rigidity and durability to function as a structural distal link under repeated dynamic contacts.

E. On-Robot Morphological Reconfiguration

We finally evaluate morphological reconfiguration using the complete robot. Unlike the component-level experiments in Sec. V-B, the mechanical work required for deformation is provided here by QuadMorph's existing joint actuators.

For the wheel-to-leg transition, the robot first enters the prescribed reconfiguration pose and activates the embedded heaters. Once the shins reach the compliant state, the robot executes the reconfiguration motion described in Sec. IV-D, using contact with the ground to reshape the softened structures. The robot maintains the target configuration during cooling before transitioning directly to the legged locomotion policy. The reverse sequence restores the wheel morphology.

Across $[N]$ trials, QuadMorph successfully completes the wheel-to-leg transition in $[\cdot]/[N]$ trials and the leg-to-wheel transition in $[\cdot]/[N]$ trials. The average reconfiguration time is $[\cdot]$ s, with an electrical energy consumption of $[\cdot]$ J per shin.

F. Multimodal Locomotion

To demonstrate the benefit of physical reconfiguration at the system level, we construct a mixed-terrain course containing regions suited to different locomotion morphologies. The course consists of a smooth rolling section, followed by [rough/discontinuous terrain or obstacles] that cannot be reliably traversed in wheel mode, and a second smooth section.

QuadMorph begins in wheel mode and efficiently rolls across the first section. Upon reaching the challenging terrain, the robot stops and transforms its shins from wheels into legs using the on-robot reconfiguration procedure. It then

switches to the legged locomotion policy and traverses the [obstacle/rough-terrain] section. After returning to smooth terrain, the robot transforms back into wheel mode and resumes rolling.

The complete experiment demonstrates a continuous sequence of

$$\boxed{\text{Roll} \rightarrow \text{Morph} \rightarrow \text{Walk} \rightarrow \text{Morph} \rightarrow \text{Roll}}, \quad (13)$$

using the same distal structures throughout the task. These results demonstrate the central capability of QuadMorph: programmable material properties enable a robot to physically select between distinct locomotion morphologies according to the mobility requirements of its environment.

VI. CONCLUSION AND FUTURE WORK

This work presented **QuadMorph**, a quadrupedal robot with material-enabled reconfigurable shins for wheeled and legged locomotion. By integrating a custom 3D-printed thermally responsive polymer with embedded liquid-metal Joule heating, the proposed shin can transition between a circular wheel geometry and an elongated leg geometry. Rather than relying on a dedicated rigid-body transformation mechanism, QuadMorph temporarily softens its load-bearing structure and reuses the robot's existing joint actuators and environmental contacts to reshape its morphology. Once cooled, the structure recovers its stiffness and maintains the new configuration without continuous actuation.

We demonstrated this concept through material and component characterization, on-robot morphological reconfiguration, and reinforcement-learning-based locomotion in both configurations. The results show that the same distal structure can support efficient rolling on smooth surfaces and legged locomotion over terrain where discrete ground contacts are advantageous. Together, these experiments demonstrate a different approach to multimodal robot design in which controllable material properties, rather than additional kinematic mechanisms, enable changes in locomotion morphology.

Several limitations remain. Thermal reconfiguration is substantially slower than conventional motor-driven mechanisms, and its performance is affected by heating and cooling dynamics, ambient temperature, and material fatigue over repeated cycles. Future work will investigate faster thermal management, improved material formulations, and shin geometries with higher load capacity and reconfiguration repeatability. We also plan to integrate terrain perception and autonomous morphology selection, allowing the robot to determine *when* and *where* to transform based on terrain geometry and locomotion efficiency. More broadly, we envision extending the proposed principle beyond wheel-leg transformation toward robotic structures whose geometry and mechanical function can be actively programmed by combining material-state control with the robot's existing actuation.

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